
Context Rot Is Three Mechanisms: Starved Evidence, Attended Impostors, Installed Precedent

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Abstract

1 Long conversations are said to wear language models down, but the standard evalu-
2 ations confound *accumulation* (more context) with *dilution* (the answer-bearing
3 tokens becoming a smaller, more distant, more contested fraction of it). We build
4 a harness that separates them: streams of individually localized tasks in which
5 context length, evidence distance, context confusability, and demonstrated answer
6 shape become four independent knobs, instrumented with a span-attention clamp
7 that can *set* any span’s post-softmax attention share. Three findings. (1) Accu-
8 mulation alone is free: accuracy and instruction adherence are flat to 90% fill,
9 with bounded nulls. (2) Displacing the evidence costs 0.19–0.21 accuracy, and
10 the cost is *causally mediated by attention mass*: removing the evidence’s share
11 with nothing moved reproduces 114% of the penalty, the mass→accuracy curve
12 is smooth with no threshold, and two separately-run clamp experiments agree
13 to 0.004. (3) Near-duplicate context costs 0.085–0.093 accuracy at *unchanged*
14 *evidence share* (−0.0003 [−0.0009, +0.0004])—but closing the context’s occur-
15 rences of the probe’s own answer candidates at generation time recovers 59% of
16 that penalty (+0.060 [+0.008, +0.112] against a size-matched random-closure
17 control at zero), so competition is attention-mediated too, through the *competitors*
18 rather than the evidence. (4) Accumulated filler that demonstrates an *applicable*
19 answer shape erodes an instructed output format totally (0.875 → 0.000 by fill 0.15,
20 accuracy unharmed) while the instruction’s per-token attention is flat-to-rising and
21 its content stays perfectly linearly decodable. Attention mass is nonetheless a causal
22 *weight* in that contest: clamping the instruction’s share down collapses compliance
23 where nothing competes, clamping it up restores it against 42 counter-exemplars,
24 and restating the policy restores it without surgery—and the same closure that
25 rescues competition restores nothing here: the mode is installed at prefill. Context
26 rot, as measured, is not a decay of competence with length: it is displacement (the
27 evidence starved of attention), competition (attended impostors), and precedent (an
28 installed mode)—attention dilution proper is only the first.

29 1 Introduction

30 A recurring worry about long conversations is *context rot*: as turns accumulate, the model is said
31 to degrade. The phenomenology is real. On a repeating multiple-choice task, instruction-tuned
32 models become dramatically more confident as context fills—next-token entropy drops 3–4× on
33 Qwen-2.5-7B and Llama-3.1-8B, in both question orders—and the last token’s attention shifts
34 systematically away from the system prompt and the current query toward the most recent prior cases,
35 with the distribution diffusing (at layer 24 of OLMo-2-Instruct: system↔fill −0.93, recent↔fill
36 +0.89, current↔fill −0.71, entropy↔fill +0.95, all monotone in fill). By every visual diagnostic this
37 is a strong effect that grows with context.

The tempting reading is that the model is being worn down by length. But almost every context-rot evaluation embeds the answer-bearing information inside *one long task*—a long document, a buried needle, a growing codebase. There, longer context mechanically *dilutes* the relevant tokens among irrelevant ones, so a measured degradation conflates the cost of accumulation per se with the cost of the target being harder to attend to (“lost in the middle,” Liu et al., 2023). These cannot be told apart in that setup, and the distinction matters practically: an agent transcript of finished, locally-scoped steps accumulates without diluting, and the two accounts make opposite predictions about it. (Appendix A places this work in the literature.)

We separate the candidate causes inside one harness (Fig. 1) and ask three questions:

RQ1 Does accumulation alone—more context, with the evidence for every answer held at the current query—degrade accuracy or instruction adherence?

RQ2 When dilution does cost accuracy, is the cost mediated by the evidence’s attention mass? We split dilution into its two components—the evidence becoming *distant* and the context becoming *confusable*—and test each causally.

RQ3 What erodes instruction-following under accumulation, if (§4.1) adherence survives context fill? Dongre et al. (2026) recently showed that attention from responses to the system prompt closes over accumulated filler and asked whether behavior survives that closure, finding the answer architecture-indexed; our precedent arms take up their question with a graded instrument on one architecture.

The answers: accumulation alone is free (RQ1); distance costs accuracy through the *evidence’s* attention mass, while confusability costs accuracy at constant evidence mass through attention to the *competing instances* themselves—closing them recovers most of the penalty (RQ2); and instructed output format is lost to a *precedent context* with demonstrated answer shapes, in which the instruction’s attention mass is a causal weight but not the natural driver—so whether behavior survives attention closure is decided by *content*, by what the filler demonstrates, not by attention (RQ3).

Our contributions are as follows:

- A harness that makes accumulation and dilution independent, by accumulating individually localized tasks whose evidence never moves, and a span-attention clamp that *sets* any span’s post-softmax share with a bit-identical no-op at scale 1 (§2).
- A causal mediation result for displacement: removing the evidence’s attention mass with nothing moved reproduces 114% of the displacement penalty, and the mass→accuracy relationship is smooth and graded with no threshold in it (§4.2).
- A dissociation establishing a *second* mechanism: competition costs accuracy at constant *evidence* mass, yet closing the competitor mentions recovers 59% of the penalty—both accuracy mechanisms are attentional, on different spans, and span-targeted closure tells them apart (§4.3).
- A *third* mechanism at the compliance level: accumulated demonstrations erode an instructed format only when their shape is applicable, immediately and with accuracy unharmed; the erosion is not attention-mediated, yet re-weighting or re-presenting the instruction each fully reverse it (§4.4–§4.5).

2 Methodology

2.1 The localized-stream framework

Our harness accumulates **individually localized tasks**: a stream of self-contained 5-option medical cases (DDXPlus, Tchango et al., 2022) or discrete factual questions (MMLU, Hendrycks et al., 2021). Each answer depends only on the *current* question, which always sits at the end of the context; prior turns are never required to answer it. Accumulation therefore grows the context without ever diluting the information the current answer needs, and dilution becomes a set of *independent knobs* (Fig. 1): holding the item, the model, the total fill and the metric fixed, we can move the evidence k user turns back (*displacement*), replace the filler with near-duplicate cases (*competition*), or let the filler’s replies demonstrate a competing answer shape (*precedent*)—with question text byte-identical across arms and the same filler in every arm unless the filler is the treatment. Distance, fill, confusability, and demonstration, welded together in a long-document setup, move independently here.

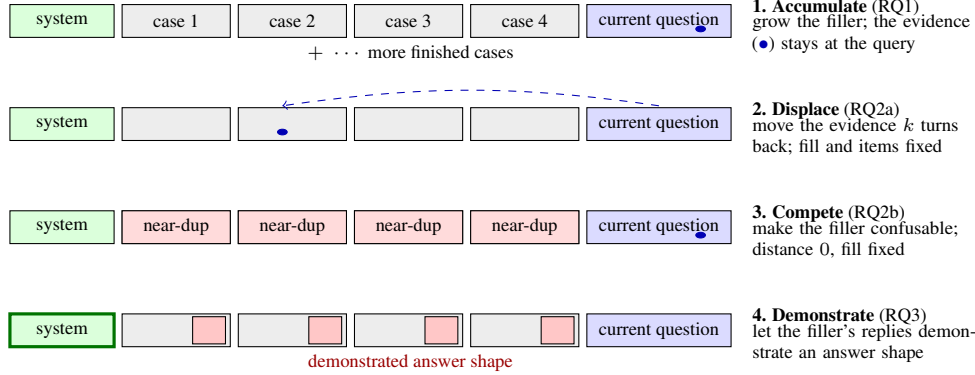


Figure 1: **The localized-stream framework: four independent knobs on one transcript.** Each stream accumulates self-contained cases, so the evidence for the current answer sits at the current question by default and every knob is turned with the others held fixed. The span-attention clamp (§2.3) additionally sets any span’s attention share directly, turning the mediating variable itself.

90 2.2 Span share and enrichment

91 For a token span A (the evidence, the system prompt, a filler answer), its *share* is the final position’s
 92 post-softmax attention mass on A , averaged over heads at a reference layer (layer 24 unless stated) or
 93 over all 32 layers where noted. A fixed-size span’s raw share falls mechanically as context grows—a
 94 48-token system prompt is a shrinking fraction of the key space—so raw share declines say nothing by
 95 themselves. We therefore also report *enrichment*: share divided by the span’s token fraction, which is
 96 1 for a span attended in proportion to its size. The distinction does real work in §4.4: raw system share
 97 falls $\sim 10\times$ on the same curve in arms that erode and arms that do not, while enrichment separates
 98 nothing—and metrics that aggregate raw attention-to-goal share across a growing context, such as
 99 the goal-accessibility ratio of Dongre et al. (2026), build this arithmetic decline in by construction.

100 2.3 The span-attention clamp

101 To intervene on attention rather than position, we use a clamp that *sets* a span’s post-softmax share: a
 102 constant bias b added to the span’s key columns in the additive attention mask scales the span’s odds
 103 by exactly e^b , and softmax renormalizes. Nothing in the attention forward is reimplemented, so a
 104 scale of 1.0 ($b=0$) is bit-identical to the unhooked model; a bisection on b hits any target share at the
 105 reference layer to $\sim 10^{-4}$; and scale 0 closes the span outright—the hard-masking intervention of
 106 Dongre et al. (2026), recovered as the clamp’s endpoint, so their ablation and our graded doses live
 107 on one dial. The clamp is graded and bidirectional—it can remove a span’s mass or restore it—which
 108 is what lets us test mediation in both directions rather than only ablate. It is uniform across heads
 109 and, unless stated, applied at every layer; per-head structure is measured separately (Appendix E).
 110 Unlike attention-steering methods aimed at improving compliance (Zhang et al., 2024), the clamp
 111 is a measurement instrument: its no-op is exact, its dose is solvable, and its near-ablation regime is
 112 excluded from interpretation (§4.2). Validation is in Appendix B.

113 2.4 Residual probes and mode-vector steering

114 To ask what the residual stream represents while behavior changes, we train linear probes (Standard-
 115 Scaler $\rightarrow$ PCA(≤ 50) $\rightarrow$ LDA) on the final pre-generation position across all 33 stack rows, with
 116 leave-one-out cross-validation at episode level and 200-shuffle permutation nulls at peak layers—a
 117 protocol comparable to recent residual-probing work on multi-turn degradation (Dongre et al., 2026).
 118 To test what a decodable direction *does*, we steer: mean-difference vectors added at the final prefill
 119 position and every decode step (never the context), always against a norm-matched random-vector
 120 control at the same dose, and projection-erasure of fitted directions at every position and step. Details
 121 in Appendix G.

3 Experimental design

Streams and probes. All causal experiments run on OLMo-2-7B-Instruct at seed 42, greedy decoding. A 4–32k window holds only tens of cases, so we pool many independent accumulation sessions; a reversed-question-order control removes difficulty-ordering confounds. Probe options are shuffled per item and items with fewer than 5 options are excluded (DDXPlus lists differentials in rank order, so unshuffled gold letters are 71% “A”—a letter-bias trap for any arm comparison; Appendix C).

Compliance instrumentation. For RQ3 the system prompt carries a clinical answer template (ANSWER: letter, then SUPPORTING: with at least two vignette findings) and every reply is graded separately for compliance and accuracy by a rule-based checker; simpler canary instructions (a reply prefix and suffix) instrument adherence in the accumulation and clamp arms. Grader rules, filler construction, and the three filler streams’ demonstrated shapes are specified in Appendix C.

Statistics. Per-condition n is bounded by the context window, so every clamp experiment is paired over shared items and analysed with a paired bootstrap resampling *cases* (10,000 draws), never heads or turns; resampling arms independently inflates intervals about $2.5\times$. An overflow guard skips rather than truncates every item that would not fit whole—truncating a long item near the window edge manufactures exactly the late-window cost under study—and reports per-arm skip counts (Appendix D).

4 Results

4.1 Accumulation is free

Per-case accuracy does not fall as context fills; it is worst at the cold start and warms up ($\text{corr}(\text{correct}, \text{fill}) = +0.11$ Instruct, $+0.07$ DPO). The obvious objection is that a homogeneous repeating task lets in-context learning mask a hidden decay, so we remove that crutch: in a *random-subject* stream each question is drawn uniformly from all 56 MMLU subjects, so prior context carries no predictive information about the next question—only the answer *format* is learnable. Accuracy is still flat (random: $\text{corr} = -0.02$ $[-0.10, +0.05]$, $n=699$; coherent single-subject: $\text{corr} = +0.01$, $n=1001$). These nulls are *bounded*: the 95% interval on upper-minus-lower-half accuracy excludes declines beyond 4.1 points (coherent) and 9.4 (random). Checkable instruction adherence—the canary instructions—stays at ceiling throughout ($\text{corr}(\text{violation}, \text{fill}) = 0$), an honest null that bounds rather than proves, since the canaries were easy; §4.4 shows what it does and does not mean. Accumulation does drive the *current query*’s attention share down to exactly the shoulder at which clamping the same share starts to cost points (Appendix F)—that accuracy nonetheless stays flat is itself evidence that in-context learning is actively holding it up.

4.2 Displacement acts through attention mass

We now vary only *where the answer-bearing evidence lives*. Each DDXPlus probe’s vignette is placed k user turns before its question; mean fill is 0.688 in every arm, the question text is byte-identical, and the overflow guard skipped 0 of 192 probes. Every gap’s CI excludes zero (Table 1, Fig. 2a). In a joint fit of accuracy on fill and distance, **distance carries the coefficient** ($\beta = -0.00761$ $[-0.01173, -0.00346]$, significant) and **fill carries none** ($\beta = -0.00725$ $[-0.21006, +0.18658]$). Within the *local* arm—the one that reproduces the accumulation design—accuracy is flat with fill ($\beta = -0.294$ $[-0.767, +0.184]$). The effect survives restriction to parsed responses and is then fully monotone ($0.524 \rightarrow 0.413 \rightarrow 0.397 \rightarrow 0.343 \rightarrow 0.338$). One honest qualification: the decline saturates by $k \approx 10$ rather than deepening to $k=20$, and those two arms are not distinguishable from each other. So the same model, on the same items at the same fill, degrades when the evidence is displaced and does not degrade when it is not. The localized null is not a ceiling artifact: these items *can* show degradation, and do, as soon as dilution is reintroduced.

Distance and attention drain move together by construction, so Table 1 is consistent with dilution but does not establish it—a purely positional account predicts the same table. Four interventions separate them.

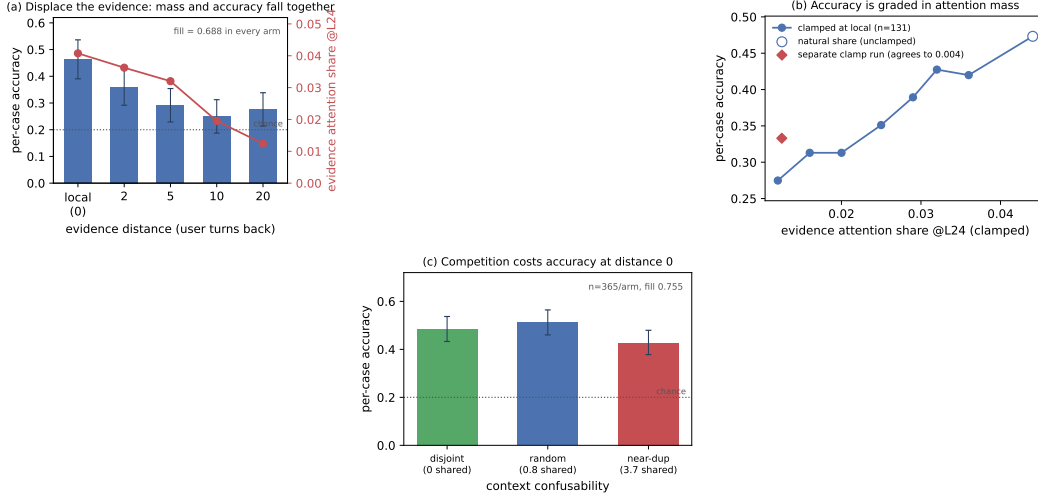


Figure 2: **Dilution costs accuracy; accumulation does not.** (a) Moving the evidence k user turns back, at identical fill and byte-identical questions, drains its attention mass and costs accuracy together. (b) Setting that mass directly, with the evidence left in place, reproduces the penalty and traces a smooth graded dose-response—no threshold. The diamond is an independently-run clamp experiment measuring the same endpoint cut; the two agree to 0.004. (c) At distance 0 and fixed fill, near-duplicate accumulated context costs a further 0.085. Bootstrap CIs resample cases throughout.

Table 1: **Distance is the variable that matters** (OLMo-2-Instruct, $n=192$ per arm, chance = 0.200, fill 0.688 in every arm). Evidence attention share is measured at layer 24 on the same forwards. Case-resampled bootstrap 95% CIs.

arm	local	back_2	back_5	back_10	back_20
accuracy	0.464	0.359	0.292	0.250	0.276
cost vs local	—	+0.104	+0.172	+0.214	+0.188
95% CI	—	[+.005, +.203]	[+.073, +.266]	[+.120, +.307]	[+.094, +.281]
evidence share @L24	0.0408	0.0363	0.0320	0.0195	0.0124

Observational correlations. Evidence share falls $0.0408 \rightarrow 0.0124$ across the ladder ($r = -0.83$ with distance) while the *question*’s share stays flat, confirming only the evidence moved. A trap worth recording: *within* an arm, higher evidence share predicts *lower* accuracy ($\beta = -11.2 [-20.0, -2.6]$, stronger controlling for vignette length). Attention there tracks item difficulty—the model looks harder at confusing vignettes—and has the *opposite sign* to the causal effect below. Anyone reading share-against-accuracy correlations off this data without intervening will conclude the reverse of the truth.

Mass removal. Keeping the evidence at local and clamping its share down to the *same item*’s own back_20 share costs +0.202 [+0.138, +0.267] (paired $n=174$) and lands at 0.333 against back_20’s 0.365—a difference of $-0.025 [-0.083, +0.035]$, indistinguishable. **Mass removal alone reproduces 114% of the displacement penalty, with nothing moved.**

Dose-response. Sweeping the evidence’s share at fixed position over seven levels (common subset $n=131$ present at every level) gives a monotone decline from 0.473 at the natural share (0.0441) to 0.275 at 0.012 (Fig. 2b). Six of the seven levels differ significantly from the natural share—including a cut of less than a fifth of the mass—and no step is a threshold: the largest adjacent step is 0.053 and the decline is smooth throughout. The endpoint contrast, +0.198 [+0.115, +0.282], matches the independently-run clamp experiment’s +0.202 [+0.138, +0.267] for the same share change—agreement to 0.004 across two separate runs, which is the strongest internal consistency check in the program. An earlier draft of this work read a *threshold* into two flat contrasts; the sweep refutes it. The relationship is *graded*: a measured slope, not a cliff.

191 **Mass restoration.** The converse is only partial: clamping the displaced evidence’s share back
 192 *up* to its local level—share-matched across all 32 layers, not indexed to one—recovers a real
 193 but small 0.28 [0.07, 0.61] of the penalty (+0.047 [+0.010, +0.083] of +0.167 [+0.099, +0.240]),
 194 leaving most of it in place. Our uniform clamp can strip mass faithfully but cannot reconstruct *which*
 195 heads should carry the restored mass. Full sufficiency with partial necessity is the signature of an
 196 instrument that is faithful in one direction and crude in the other, and we report it as one rather than
 197 as a second mechanism. A pattern-matched, per-head clamp is the test that would settle it. The drain
 198 itself is positional in *tokens*, not turns—a 64% mass cut from token distance alone costs at most 9.4
 199 points—consistent with RoPE decay rather than conversation structure (Appendix F).

200 4.3 Competition acts through the competitors, at constant evidence mass

201 Burying a needle varies two things at once: the evidence becomes distant *and* it becomes surrounded
 202 by plausible alternatives. We isolate the other half by holding the evidence at the current query and fill
 203 fixed, and varying only how *confusable* the accumulated context is. Every arm accumulates eight prior
 204 DDXPlus cases, each answered in the transcript with its gold letter, so format and in-context-learning
 205 affordance are constant; the arms differ only in how much the accumulated cases’ differentials overlap
 206 the current probe’s (disjoint 0 shared options, random 0.80, near_dup 3.75 of 5, always with a
 207 *different* gold, so the context never displays the probe’s answer as a correct one). Paired over $n=365$
 208 probes, accuracy is random 0.512, disjoint 0.485, near_dup **0.427** (Fig. 2c).

209 Near-duplicate context costs -0.085 [$-0.140, -0.030$] against the natural stream and -0.058
 210 [$-0.112, -0.006$] against disjoint context; in a joint fit within this experiment, **option overlap**
 211 **carries the coefficient** ($\beta = -0.021$ [$-0.039, -0.003$]) and **fill carries none** ($\beta = -0.284$
 212 [$-0.640, +0.075$])—the same shape as the distance result, on a different variable. Two controls
 213 matter: the low-competition arms must land on §4.2’s local accuracy of 0.464, and both do (+0.049
 214 [$-0.037, +0.134$] and +0.021 [$-0.065, +0.107$]); and context length does not order the arms, since
 215 disjoint is the *longest* and mid-accuracy while near_dup is the shortest, so length biases against
 216 the result. Restricted to parsed responses only, random – near_dup = +0.082 [+0.025, +0.138]
 217 survives. The effect is *not* monotone in confusability: random sits above disjoint, not below it,
 218 though not significantly (+0.027 [$-0.019, +0.074$]). We therefore claim that near-duplicate context
 219 is costly, *not* that cost rises with overlap—one significant step, not a gradient.

220 **The dissociation on mass.** Repeating the sweep with attention capture (same probes, same seeds)
 221 and measuring every one of the model’s 32 layers separates the two mechanisms on mass. Averaged
 222 over all layers, displacement removes 0.0455 of the evidence’s attention share and competition
 223 removes 0.0022, while costing 0.188 and 0.085 accuracy respectively. Competition is therefore
 224 far cheaper in mass per point of accuracy than displacement—but it is *not* mass-neutral, and the
 225 layer chosen decides how it looks: at layer 24 the headline contrast moves the evidence’s share
 226 by -0.00027 [$-0.00088, +0.00035$], indistinguishable from zero, while at layer 17 it moves it by
 227 -0.0186 . The two drain profiles are *negatively* correlated across layers ($r = -0.32$): displacement
 228 bites hardest at layers 0–3 and 16, competition at 17–18. That the same accuracy contrast has
 229 opposite attention signatures in different layers is evidence for two mechanisms which no single-layer
 230 measurement could supply. Per-head structure—including the null result that no head hides a large
 231 countervailing move—is in Appendix E.

232 **The competitor-closure test.** Constant evidence mass rules out starvation; it does not rule out the
 233 other attentional route, in which the filler’s (arm-constant) mass lands on *instances of the probe’s*
 234 *own answer candidates* and reading them is what misleads. We test it directly: closing every
 235 context occurrence of the probe’s option names at generation time (scale 0; 30.0 spans, 127.9 tokens,
 236 union share 0.0077 per probe) recovers **59%** of the competition penalty (+0.055 [+0.006, +0.104],
 237 paired $n=365$; +0.060 [+0.008, +0.112] against a size-matched random-closure control that lands
 238 at zero, -0.006 [$-0.036, +0.025$]), and the residual gap to random no longer excludes zero (+0.038
 239 [$-0.011, +0.088$]). The rescue survives restriction to parsed responses (+0.058 [+0.003, +0.113]).
 240 **So competition is attention-mediated after all—through the competitors, not the evidence:** less
 241 than 1% of the attention distribution carries the effect, and removing it removes most of the cost. The
 242 claim that survives is sharper than the one it replaces: displacement is the evidence *starved* of mass,
 243 competition is impostors *attended*, and the evidence-mass measurement dissociates them because it is

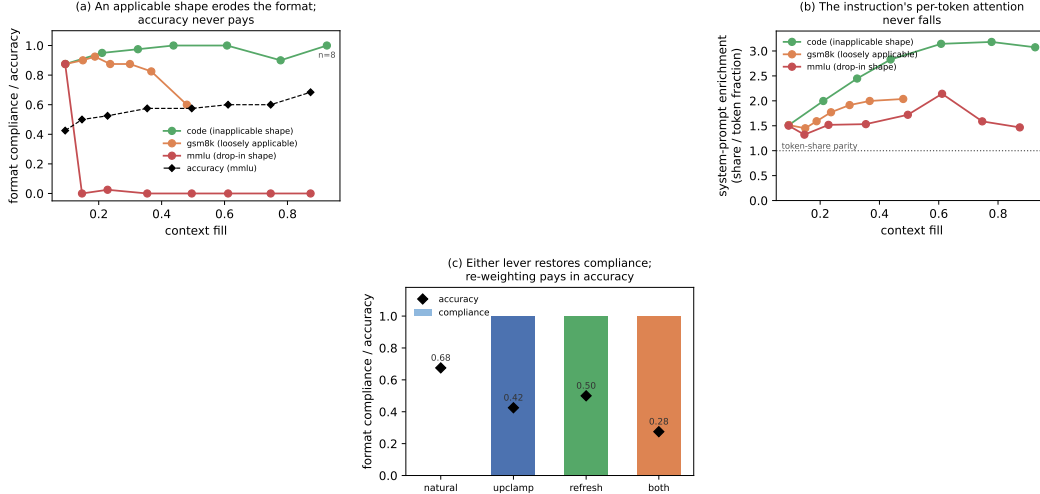


Figure 3: **Format erosion is a precedent contest, not attention starvation.** (a) Compliance by fill for three filler streams whose demonstrated answer shapes differ only in applicability to the probe; mmlu accuracy (dashed) is unharmed through its total collapse. (b) The instruction’s per-token enrichment is flat-to-rising in every arm, collapse included. (c) At depth 42, clamping the system span’s share back up and restating the policy each restore compliance fully; forced mass is paid for in accuracy (zero-sum), restatement is nearly free.

blind to the second by construction. What remains open is whether the $\sim 40\%$ residual is prefill-borne interference or instrument slack (closure covers verbatim option-name mentions only).

4.4 Precedent: instructions lose to applicable demonstrations

The costs so far are to accuracy. A third mechanism costs *compliance*: not whether the model can still solve the task, but whether it answers in the instructed form (§3; $n=40$ probes per depth, one accumulation snapshotted at every depth, so depth is the only mover).

The instruction’s attention is causally necessary—and demonstration masks it. Accumulation drives the system span’s share down $8\times$ within eight turns ($0.166 \rightarrow 0.021$). Clamping it from 0.165 to 0.050 —a level accumulation passes on its own between two and four turns, at -2.6 nats, well clear of the near-ablation regime excluded in §4.2—in a context containing no assistant turn collapses compliance from 0.99 to 0.03 (a suffix canary flips on 120 of 120 items) while accuracy moves $0.525 \rightarrow 0.467$ ($[-0.017, +0.133]$) at parse rate 0.967 : compliance is destroyed and competence is not. This confirms Dongre et al. (2026)’s ablation finding that system-prompt attention is causally necessary—and sharpens it: the collapse needs no masking, only a graded clamp to a share accumulation reaches by itself. But one *compliant* example in context pins compliance at ceiling at every clamp level, and one bare-letter example pins it at floor even unclamped—replies byte-copy the previous assistant turn’s format, with zero length variance across 720 generations per arm (Appendix G). The instruction governs behaviour only when the transcript is silent, which re-reads §4.1’s adherence null: that null accumulates *compliant* history, so it shows compliance survives compliant precedent, not accumulation.

Erosion tracks applicability, not attention. Three filler streams differ in what their answers demonstrate: code (request $\rightarrow$ Python function; no shape applicable to a 5-option probe), gsm8k (worked prose; loosely applicable), mmlu (MCQ $\rightarrow$ bare letter; a drop-in replacement). code never erodes (compliance 0.875 at cold start, 1.00 at matched fill 0.5 , 0.90 at fill 0.78); gsm8k erodes late and partially (0.600 at fill 0.48 , replies drifting into the filler’s worked-solution register); mmlu collapses totally and immediately, $0.875 \rightarrow 0.000$ at three filler turns (fill 0.147), never recovering through fill 0.87 (Fig. 3a). At matched fill ≈ 0.5 the arms order $1.00/0.60/0.00$. Accuracy on the same replies is unharmed everywhere (0.425 shallow, 0.500 – 0.684 deep): the model still solves the case—it answers in the demonstrated style. The system prompt’s attention does not carry the difference: its raw share falls on the same arithmetic curve in every arm, its per-token enrichment

is flat-to-rising in all three (Fig. 3b), and code holds full compliance at half the raw share at which mmlu has already fully collapsed. This answers the survival question of Dongre et al. (2026) on an axis their fixed-filler design cannot see: the same model at the same near-zero share survives or collapses depending on what the filler demonstrates.

Reversal is total, by either lever. At depth 42 (natural compliance 0.000, accuracy 0.675), clamping the system span back to its cold-start share restores compliance to 1.000 at an accuracy cost of $0.675 \rightarrow 0.425$ —mass forced onto a 48-token span is taken from everything else, §4.2’s zero-sum arithmetic—and restating the policy in the latest user turn also restores 1.000, at 0.500 and with no surgery (both at once: 1.000 at 0.275; Fig. 3c). Natural dilution therefore does not cause the erosion—it is identical in the arm that never erodes; an applicable competing template does—and the outcome is a contest in which attention mass is a causal *weight*: clamp the instruction down and it loses unopposed, clamp it up and it wins against 42 counter-exemplars, leave it natural and the demonstrated template beats it. Re-presenting beats re-weighting, though both work.

4.5 The mode: readable, installable, not erasable

Where is the demonstrated format, if not in attention? The demonstrated answer spans *are* preferentially read exactly where erosion occurs (per-token enrichment of filler answers over filler questions: +1.28 in mmlu, +0.34 in gsm8k, negative in code—the erosion ordering), but reading them is not what sustains it: masking every exemplar answer’s key columns at generation time restores nothing (0.000 compliant; a size-matched random-masking control is equally null), so the privileged reading is an epiphenomenon of the template’s installation at prefill, task-vector style (Hendel et al., 2023). The contrast with §4.3 is the sharpest dissociation in the paper: the *same* scale-0 closure that recovers 59% of the competition penalty restores nothing here. Competition is generation-time *reading*; precedent is prefill *installation*. One intervention, opposite outcomes, two mechanisms.

Residual-stream probes (§2.4) complete a three-level dissociation: the instruction’s presence remains perfectly linearly decodable at every depth (transfer AUC 1.000, permutation $p=0.000$) while it has zero behavioral force—read, represented, overruled—and whether the upcoming reply will comply is decodable *before the first token* (leave-one-out AUC 0.822 at layer 21) within cells matched on depth, fill and filler. The mode is installable by direction: adding the mean-difference vector at decode time to a demonstration-free context destroys the instructed format (0.000 vs. 0.525 under a norm-matched random vector), the replies reproducing the demonstrated registers verbatim. But no linear direction we tried *erases* it—mean-difference at one layer and at eleven, and the probe’s own reader direction, all with clean controls (Appendix G)—even though two fitted projections drive the layer’s linear code to chance: the code is low-rank, not redundant, so the behavioral decision does not consume the linearly decodable component.

This asymmetry carries a methodological caution beyond this paper. Behavior modes are increasingly localized in compact linear structure—rank-1 adapters that capture reasoning behavior with interpretable directions (Ward et al., 2025), task and function vectors (Hendel et al., 2023; Todd et al., 2024), and near-perfect residual decodability of goal information after attention has closed (Dongre et al., 2026). Our mode is exactly such an object on the read and write sides—and still no readable direction is its carrier. Decodability and installability licence no inference to necessity: the probe reads a shadow of the mode, not the mode.

4.6 Why the folklore exists: the signatures

If the entropy collapse behind the context-rot folklore were a generic depth effect it should survive on heterogeneous dialogue. It does not. On 150 organic WildChat conversations (Zhao et al., 2024) (Qwen-2.5-7B, teacher-forced over the real history), within-conversation entropy is flat with depth: median late/early ratio 0.99, and 52% of conversations trend down—a coin flip. Partitioning WildChat by its *own* inter-turn homogeneity isolates the driver: homogeneous conversations collapse (0.897 late/early) while heterogeneous ones are flat (1.001), the entropy slope correlates with homogeneity (partial $r = -0.151$ controlling for length), and because homogeneous conversations are *longer*, length works against the effect (Fig. 4b). Across OLMo-2’s (OLMo Team, 2024) post-training chain the collapse is mostly a downward *level shift* in baseline entropy installed by alignment ($1.06 \rightarrow 0.20$ across base→SFT→DPO→Instruct), not a steeper collapse slope (Appendix H).

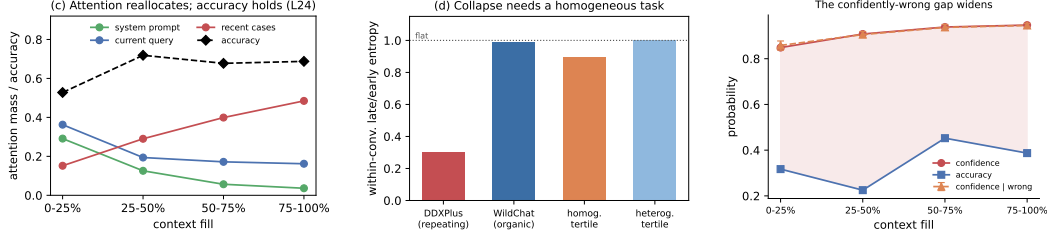


Figure 4: **The signatures are in-context comfort, and the residual hazard is calibration.** (a) Attention reallocates monotonically off the system prompt and current query with fill while accuracy holds. (b) The entropy collapse appears only with a homogeneous repeating task, not on organic WildChat dialogue, and tracks homogeneity rather than length. (c) Confidence rises with fill on wrong answers as much as right, at flat accuracy.

The residual hazard in the localized case is *calibration*, not competence: on the DDXPlus streams logging per-token confidence ($n=154$ pooled cases, two models, both question orders), answer confidence rises $0.85 \rightarrow 0.95$ with fill ($r = +0.72 [+0.64, +0.79]$) at flat accuracy—and equally on *wrong* answers ($0.86 \rightarrow 0.95$; $r = +0.69 [+0.57, +0.78]$)—so the confidently-wrong gap widens by ~ 10 points over a session (Fig. 4c), most consequential exactly where a homogeneous high-stakes stream makes the model surest.

5 Conclusion

With accumulation, displacement, competition and precedent separated inside one harness: accumulation produces a real, monotone reallocation of attention and a real collapse of *confidence*, but no loss of accuracy and no loss of instruction adherence. Displacing the evidence produces a large accuracy cost that is causally mediated by its attention mass, graded in that mass, and positional in tokens. Making the context confusable produces a further cost at constant evidence mass that is carried instead by attention to the competing instances themselves—under 1% of the distribution, and closing it recovers most of the cost. Accumulating an applicable demonstrated answer shape erodes not accuracy but *compliance*, in a precedent contest the instruction loses with its attention intact—and which either re-weighting or re-presenting the instruction wins back completely. The literature’s central claim is therefore *correct but misattributed, and under-counted*: context rot is not a decay with length, it is displacement, competition and precedent—starved evidence, attended impostors, an installed mode—three mechanisms with different signatures, of which length is merely the usual way of causing them at once, and of which only the first is attention dilution proper. The test we would run next is the pattern-matched, per-head clamp that would settle whether displacement’s partial necessity is an instrument limitation or a second channel within displacement itself.

6 Limitations

Per-condition n is bounded by the context window, so every interval here depends on the paired analysis of §3; resampling the arms independently inflates each one about $2.5\times$, enough to report a real effect as null. The necessity direction of the mediation is only partially established and is plausibly instrument-limited, since the clamp is uniform across heads. The competition result is one significant step rather than a gradient—mild overlap is indistinguishable from none, and we do not claim cost rises with confusability. The competitor-closure rescue is substantial, not total: the residual is not distinguishable from zero but its point estimate leaves $\sim 40\%$ of the gap unexplained, closure covers verbatim option-name mentions only, and the parsed-only contrast against the random-closure control sits exactly at the interval boundary. The instruction-canary null is a floor effect, the WildChat figures are summary values from teacher-forced analyses, and the causal program runs on one model family. The precedent results are one instruction family, graded by a rule-based checker at $n=40$ per cell; and the carrier of the demonstrated-format mode is unidentified—it is decodable and installable, but no linear direction we tried erases it, so we localize the decision neither to a layer nor to a direction. The accuracy null holds for localized tasks; genuinely length-dependent single-task settings are where dilution is the point, and are out of scope by design.

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A Background and related work

Length as the suspected cause. Most evidence for context rot comes from setups where the answer-bearing information sits inside a single long input. Liu et al. (2023) showed that accuracy depends strongly on *where* in the context the relevant passage sits, dipping when it falls in the middle. Levy et al. (2024) held a reasoning task fixed and padded the input with task-irrelevant text, finding degradation well before the models’ advertised context limits. Hong et al. (2025) ran a broad sweep over 18 models and found that reliability falls with input length even on tasks as simple as retrieval and replication, and that the fall is not uniform: it depends on how similar the needle is to the question, and on whether distractors are present. Laban et al. (2025) report a large multi-turn degradation, but locate its cause in the model committing early to a wrong reading and failing to recover—not in length as such.

Confusable context as a separate suspect. A second literature manipulates the *content* of the surrounding context rather than its length. Shi et al. (2023) insert an irrelevant sentence into grade-school math problems and observe a sharp accuracy drop. In retrieval-augmented generation, Cuconasu et al. (2024) find that documents retrieved as relevant but not containing the answer act as distractors and degrade accuracy. Wang and Sun (2025) show accuracy declining log-linearly as interfering similar updates accumulate, independent of raw context length—the nearest neighbor to our competition result, with no attention instrumentation; our contribution is that this cost runs at held-constant attention mass, which rules out attention capture as its mechanism.

Instructions versus demonstrations. Demonstrations can override a model’s trained priors (Wei et al., 2023) and, at scale, its safety training (Anil et al., 2024); the same lever is framed as pure benefit in many-shot prompting (Agarwal et al., 2024). Hardcoded assistant turns exhibiting a competing pattern flip instruction-following into pattern-following (Camassa and Shiller, 2026), system-prompt adherence fails under conflicting or adversarial user turns (Mu et al., 2025), and instruction-derived and demonstration-derived task representations are measurably distinct (Davidson et al., 2025). Our eroding pressure differs from all of these: it is benign, self-generated, and contains no conflicting directive anywhere—precedent, not conflict or attack—and we identify the contest’s causal weight and both of its reversals.

Attention as measurement and lever. Closest to our mechanistic question, Dongre et al. (2026) track attention from responses to the system prompt declining over 50 filler turns across four model families, show that hard-masking the instruction out of the window collapses compliance, and decode goal information from the residual stream at high AUC after attention has “closed”. Three differences matter. Their goal-accessibility metric aggregates raw share, which falls by arithmetic as context grows (§2.2); our token-normalized enrichment removes that decline entirely, and the corrected quantity is flat in eroding and non-eroding arms alike. Their causal arm is total ablation—attention necessary, which our system-span clamp confirms—while our clamp is graded and bidirectional, and the restoration direction (compliance recovered against 42 counter-exemplars) is an arm ablation cannot supply. And their filler is always inapplicable; our applicability ladder shows the same model at the same near-zero share surviving or collapsing depending on what the filler demonstrates, a content dependence invisible to fixed-filler designs. On the steering side, Zhang et al. (2024) boost attention on profiled head subsets to improve compliance and Hsieh et al. (2024) recalibrate a static positional attention bias globally; our clamp is the measurement-grade counterpart—bit-identical no-op, solvable dose, span-targeted in accumulating multi-turn context—used to establish when attention mass is and is not the mediating variable.

Compact task representations. In-context demonstrations compress into residual task vectors (Hendel et al., 2023) transported by identifiable heads (Todd et al., 2024), and reasoning capability elicited by finetuning can be captured in a rank-1 adapter whose directions are interpretable (Ward et al., 2025). These are the no-attention-signature channel our precedent mechanism appears to use (§4.5), and our erase-null results bound what such compact readable structure can be assumed to do causally.

B Instrument validation

Attention capture. Shares are read from a reconstruction of each layer’s attention at the final position (QK-norm where the architecture applies it, then RoPE, then the additive mask, then softmax), which avoids materializing $N \times N$ attention matrices. The reconstruction matches the framework’s own output_attentions rows to max $|\Delta| = 1.5 \times 10^{-8}$ on OLMo-2 and exactly 0.0 on a GQA architecture (Qwen-2). The capture applies the additive attention mask, which is what makes it valid *under* a mask-based intervention: a capture that ignores the mask still matches on plain forwards (a causal mask’s last query row is all zeros) while silently reporting unclamped attention during a clamp.

Clamp. At scale 1.0 the clamp is bit-identical to the unhooked model (max $|\Delta|$ logits = 0.0), because it adds $b=0$ to the existing mask rather than reimplementing attention. The bias is uniform across heads, so a span’s aggregate share is hit by bisection on b ; all solved levels in this paper land within $\sim 10^{-4}$ of target. Setting a span’s scale to 0 fills its key columns with the mask minimum—

closure, the same operation as a causal mask. Interpretation is restricted to biases well clear of near-ablation: levels requiring -4.7 to -6.1 nats produce degenerate behavior (the model answers “A” 110/110) and are excluded.

C Task and filler specifications

Probe construction. DDXPlus probes are 5-option differentials with the vignette as evidence. Options are shuffled per probe; DDXPlus lists the differential in rank order, so unshuffled gold is the letter “A” in 71.4% of probes, and arms that differ in letter bias then differ in accuracy for free. Items with fewer than 5 options are excluded (26.7% of raw cases; some are answerable from a single option without the vignette). Generation runs at 256 new tokens: shorter caps truncate the reply component that formats put last, biasing exactly the metric under study.

Format instruction and grader. The clinical template requires ANSWER: with a letter and SUPPORTING: with at least two findings quoted from the vignette. Compliance requires the template; accuracy is scored separately and more permissively (a letter offered in prose, or alone on the reply’s first line, is scoreable for accuracy while remaining non-compliant). The grader was validated against real generations from every condition; graders written against imagined output formats systematically mis-score exactly the conditions where the treatment changes the output format.

Filler streams. code: 30 synthetic request→Python-function tasks on one fixed template. gsm8k: grade-school math answered in the model’s own worked prose. mmlu: 4-option MCQ (medical subjects excluded) answered with a bare letter. Filler replies are the model’s own (up to 200 tokens; mmlu answers truncated to the letter), so each arm demonstrates the shape the model itself produces for that stream. The arms are compared at matched fill, not matched depth: mmlu accumulates ~ 0.018 fill per turn against code’s ~ 0.056 .

Canaries. The adherence and system-clamp arms use a three-part canary instruction (reply prefix marker, suffix tag, and a forbidden phrase); the forbidden-phrase canary is satisfied trivially by short replies, so two canaries are informative.

D Statistical procedures

All intervals are case-resampled bootstrap 95% CIs at 10,000 draws. Clamp and arm contrasts are *paired* over shared items; the paired analysis cuts interval half-widths $\sim 2.5\times$ against independent resampling on the same data. Joint fits are linear probability models (accuracy in percentage points is the quoted scale) with case-resampled coefficient intervals. Probe analyses use leave-one-out cross-validation at episode level with 200-shuffle permutation nulls. The overflow guard skips items that would not fit whole (prompt + generation headroom) and reports per-arm counts: 0/192 in the distance sweep, 4 skipped +15 starved in competition (and the identical panel in the competitor-closure run), 2 in the mmlu erosion arm, and 32—all at the deepest depth, leaving $n=8$ there—in the code arm, whose ladder is therefore interpreted to depth 12 (fill 0.78).

E Per-layer and per-head structure

Attention share is a mean over a layer’s 32 heads, and a mean can hold still while heads cancel, so we measured all 1,024 heads under both accuracy-costing mechanisms.

Under displacement at layer 24, **every head loses evidence mass** from local to back_20, each significant at Bonferroni, so nothing is hidden by averaging. The loss is close to uniform in proportion—mean fractional drain 0.689 (sd 0.147), uncorrelated with how much mass a head held ($r=0.08$)—and a single uniform odds-scale reproduces 58% of the per-head pattern, which weakens (without eliminating) the reading of §4.2’s partial necessity as pure instrument limitation.

Under competition at the same layer the net is 0.0003 while heads move 0.0026 against a paired sign-flip null of 0.0004 ($p \leq 0.0005$, 2,000 permutations), with 19 of 32 heads individually significant: a small reallocation, about 6% of the evidence’s local share, rather than inaction. Finally, heads that concentrate on the evidence do exist—255 of 1,024 attend it more than its token share warrants, up

to 0.626 for one head on a span that is 0.102 of the context—but none of them are at layer 24, whose mean evidence share (0.0408) is among the lowest in the model (vs 0.183 at layer 3 and 0.143 at layer 16). Conclusions about head specialisation drawn from a single layer do not generalise, and we draw none.

F Additional displacement analyses

Tokens versus turns. A partial 2×2 in (gap turns) $\times$ (filler length), total context equalized by padding, dissociates them: two arms matched on *tokens* have effectively identical evidence share (0.0104 vs 0.0108) despite a $4 \times$ difference in turn count, while at matched turns share falls 0.0288 $\rightarrow$ 0.0104 as the gap grows 446 $\rightarrow$ 1684 tokens—a 64% cut in the evidence’s attention that costs at most 9.4 accuracy points (+0.026 [−0.042, +0.094]). Share regresses significantly on gap tokens and not on gap turns. Attention drain is a function of token distance, as RoPE decay predicts; conversation structure is not what drains it.

The current query’s floor. Accumulation drives the *current query*’s share from ≈ 0.35 down to ≈ 0.15 , and it is natural to ask whether that stops short of a floor. It does not. Clamping the query span on cold-start contexts, accuracy is flat from the natural share (0.258, accuracy 0.545) down through 0.20 (0.509), then costs 16.4 points at 0.15 (+0.164 [+0.036, +0.291])—which is exactly the share accumulation reaches. Accumulation does not stop far short of the shoulder; it stops *at* it. (Levels below 0.15 require −4.7 to −6.1 nats, near-ablation rather than points on a dose-response, and are excluded from interpretation.) That accuracy nonetheless stays flat under accumulation, while the same share reached by clamping costs points, is itself evidence that in-context learning is actively holding accuracy up.

G Precedent details

The system span’s natural trajectory. Measured in the clamp experiment’s own setup, the system span’s share as prior cases accumulate is 0.166 (0 cases), 0.081 (1), 0.060 (2), 0.037 (4), 0.028 (6), 0.021 (8): an $8 \times$ decline in eight turns from accumulation alone. Clamp ladders must be derived from the setup they are applied to; a share imported from a different context put two arms above natural, which were skipped rather than silently clamped upward.

One contrary example overrides the instruction entirely. With a single prior case whose answer exhibits the canaries, compliance is 3.00/3 at every clamp level down to share 0.021; with a single bare-letter answer, 1.00/3 at every level including unclamped. 720 generations per arm, zero variance in reply length (8 characters in one arm, 1 in the other): the model reproduces the previous assistant turn’s format exactly. Both arms are pinned—one at ceiling because the format is available from the transcript, one at floor because the contrary example already won—so the clamp has nothing to move, and only the neutral condition (no assistant turn anywhere) exposes the instruction’s causal necessity reported in §4.4. A “no demonstration” arm whose prior turn answers with a bare letter is not an absence of demonstration; it is a counter-demonstration.

Exemplar closure. At depth 42, closing the demonstrated answer spans’ key columns at generation time (scale 0; §2.3): all-answer closure 0.000 compliant, dose-matched partial closure 0.000, filler-question closure 0.132 (consistent with pure renormalization of the surviving spans, the system’s included), size-matched random-token closure 0.000. Accuracy moves within noise. Reading the exemplars at generation time is not what sustains the erosion.

Probes. Probe 1 (instruction presence): trained at depth 0 to separate the format system prompt from a token-length-matched neutral twin; transfer AUC 1.000 at every mmlu depth 0 $\rightarrow$ 42 (permutation nulls 0.49–0.50, $p=0.000$, $n=80/76$). The only depth trend is concentration: mean-over-layers AUC 0.985 $\rightarrow$ 0.791, peak layer drifting L1–2 $\rightarrow$ L11–14. Probe 2 (upcoming compliance): within gsm8k’s mixed cells (same depth, fill, and filler; only the outcome differs; $n=80$), LOO-AUC 0.822 at stack layer 21 (null 0.485, $p=0.000$), crossing 0.8 from layer 18. Vector geometry: the mmlu and gsm8k mode vectors (deep minus shallow mean differences) have median cosine +0.746 across layers—two textually opposite demonstrated styles shift the residual stream in largely the same direction, with the caveat that generic deep-context components are common to both.

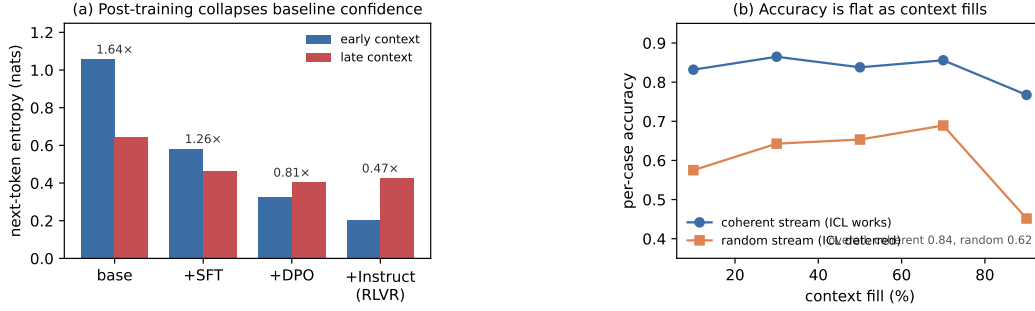


Figure 5: **Signature details.** (a) OLMo-2’s post-training chain collapses baseline confidence stage by stage; the within-context ratio *falls* across stages, so alignment installs a level shift, not a steeper slope. (b) Accuracy is flat with fill in both the coherent and the random-subject stream; the random stream is lower throughout (harder subjects), not sloped.

Steering and erasure. Install: the mean-difference vector at stack layer 21, applied at the final prefill position and every decode step at matched norm, destroys the instructed format in a demonstration-free context (0.000 vs 0.525 compliant under the norm-matched random control; natural 0.875), with replies reproducing the demonstrated registers verbatim; the collateral accuracy cost (0.175 vs 0.375) is consistent with deep-context components inside the mean-difference vector. A 3× dose breaks generation for real and random alike—controls must be matched per dose. Erase: four strategies all null with clean controls—the mean-difference direction projected out at one layer and at eleven (the eleven-layer projection actively harms while its random control is harmless), and Probe 2’s own reader direction projected out at its own layer, both in-distribution and transferred. Iterative re-probing on the captured states shows the layer-21 linear code is rank-≈2 (AUC 0.822 → 0.619 after one fitted projection, 0.505 after two), so the erase nulls are not redundancy: the behavioral decision does not consume the linearly decodable component.

H Signature details

Post-training chain. Replaying one probe across OLMo-2-7B’s base→SFT→DPO→Instruct chain (identical cases and orders) shows early-context entropy collapsing monotonically, 1.06 → 0.58 → 0.33 → 0.20, with the largest single drop at base→SFT (Fig. 5a). The “collapses as context fills” framing is mostly a base-model ICL effect plus a floor: the within-context ratio *falls* across stages (1.64 → 0.47), because aligned models start near a confidence floor.

WildChat. Median within-conversation $\text{corr}(\text{entropy}, \text{depth})$ is -0.02 on organic dialogue. The homogeneity partition (tertiles of inter-turn similarity) gives late/early ratios 0.897 (homogeneous) vs 1.001 (heterogeneous), partial $r = -0.151$ controlling for length.

A proposed mechanism that fails causally. Clamping the boundary-detector SAE feature (Gemma Scope F90871, Lieberum et al., 2024) back to its clean level on held-out fatigued cases moves entropy the *wrong* way (0.442 → 0.465 vs clean 0.293), leaves accuracy flat, and a random feature of similar magnitude perturbs entropy at least as much.

Attention and errors point the other way. Conditioning on fill, the cases OLMo-2 answers *wrong* have *higher* current-query attention than the ones it gets right (pooled $\Delta = +0.045 [-0.003, +0.097]$, $n=115$; sign positive in every fill bin and at every probed layer). Errors are associated with *fixating* on the query; correct answers spread attention onto the accumulated cases—the per-case fingerprint of the in-context learning that keeps accuracy up. The “rot” and the within-context predictor of a correct answer point in opposite directions.